

Wideband Rx

Lineup and System Specs

Original Line-up: (External LNA + Internal LNA)

System Description	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Component Name	LNA	LNA	Attenuator	Mixer	Filter
Gain (dB)	20	14	-3	-7	-2
NF (dB)	2	3	3	7	2
P1dB (dBm)	P1dB (dBm)	P1dB (dBm)	P1dB (dBm)	P1dB (dBm)	P1dB (dBm)
IP3 (dBm)	35	33	33	15	34

OUTPUTS

Parameter	Output
Gain (dB)	22.00
NF (dB)	2.04
P1dB (dBm)	-14.31
IP3 (dBm)	12.03

Original Line-up: (LPF and ADC)

LPF

Parameter	Value	Unit
0.5dB <u>VW</u>	250	MHz
3dB <u>BW</u>	600	MHz
Tunable filter	?	<u>TBD</u>
In-band IL	2	dB
OIP3	34	<u>dBm</u>

ADC

Parameter	Value	Unit
Input referred full-scale	-10	<u>dBm</u>
Resolution	10	bits
<u>F_s</u> >	2.4	GHz
<u>NSD</u>	-149	<u>dBFS/Hz</u>

Integrated noise = -149 + 10 * log(2e8) = -66dB

Original Line-up:

- Using Only Internal LNA:

at max gain

NF = 4.1270 dB

IIP3 = 10.045 dBm

Gain = 2 dB

- At min. gain:

NF = 13.406 dB

IIP3 = 18.147 dBm

Gain = -13 dB

- Using External and Internal LNA

at max gain:

NF = 2.0461 dB

IIP3_total = -9.9689 dBm

Gain = 22 dB

- At min. gain:

NF = 2.5407 dB

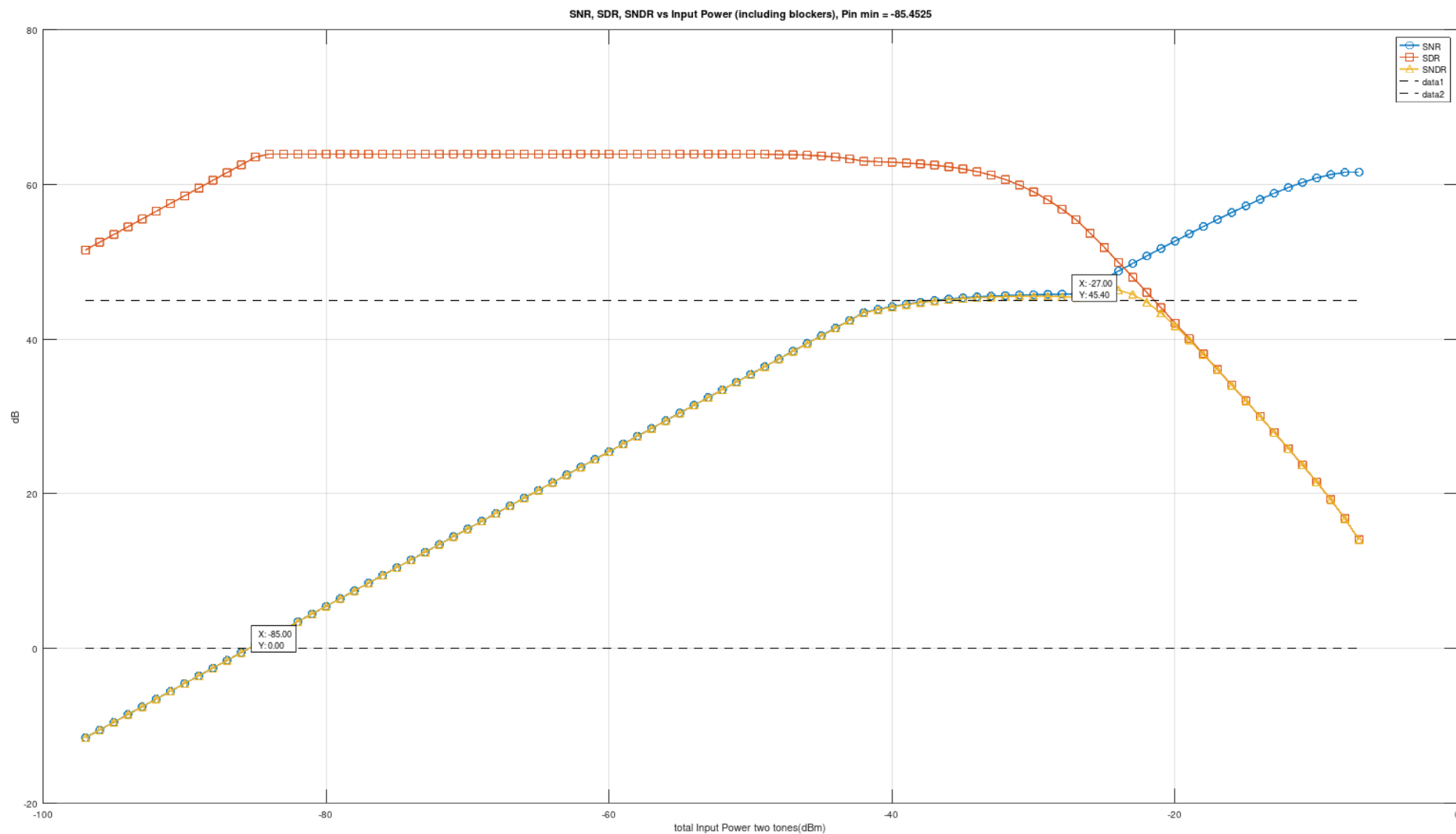
IIP3 = -1.9434 dBm

Gain = 7 dB

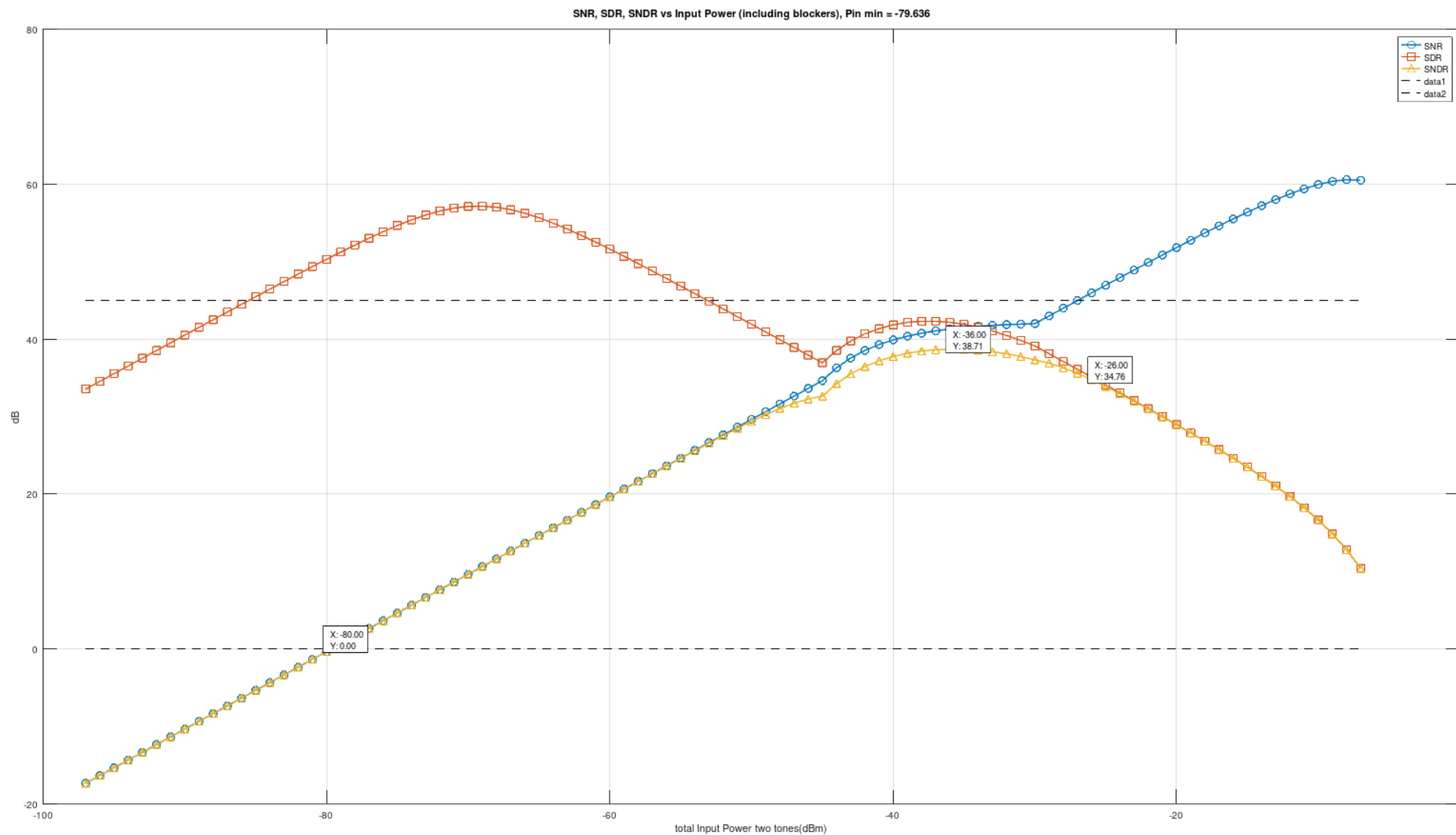
Original Line-up:

- Using Only External LNA:
at max gain
 $NF = 2.3889 \text{ dB}$
 $IIP3 = 4.2541 \text{ dBm}$
 $\text{Gain} = 8 \text{ dB}$
- At min. gain:
 $NF = 8.1867 \text{ dB}$
 $IIP3 = 13.716 \text{ dBm}$
 $\text{Gain} = -7 \text{ dB}$

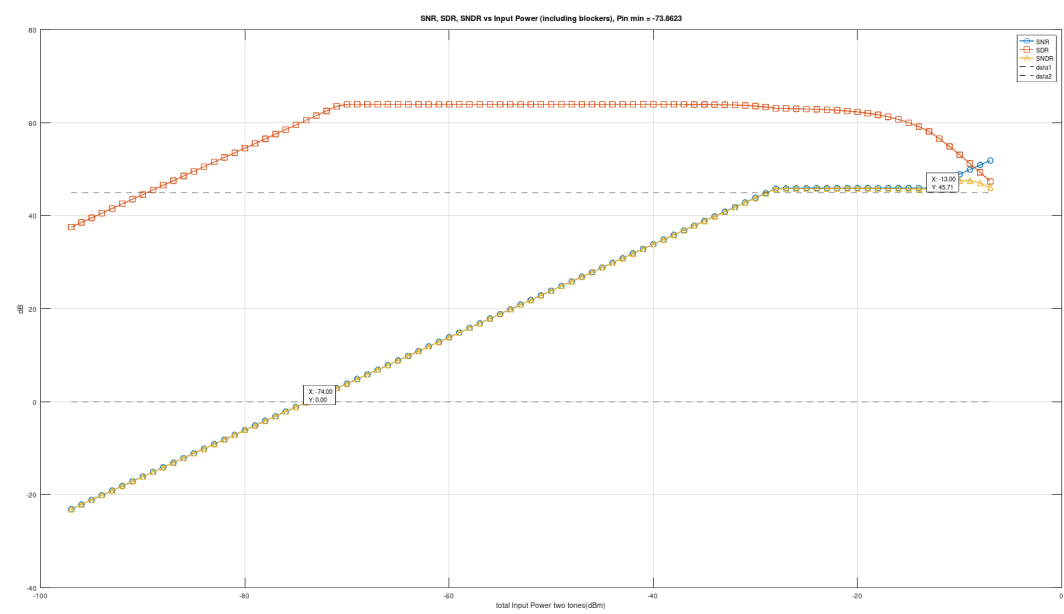
Original Lineup Results: Ex. + Int. LNAs (No Blockers)



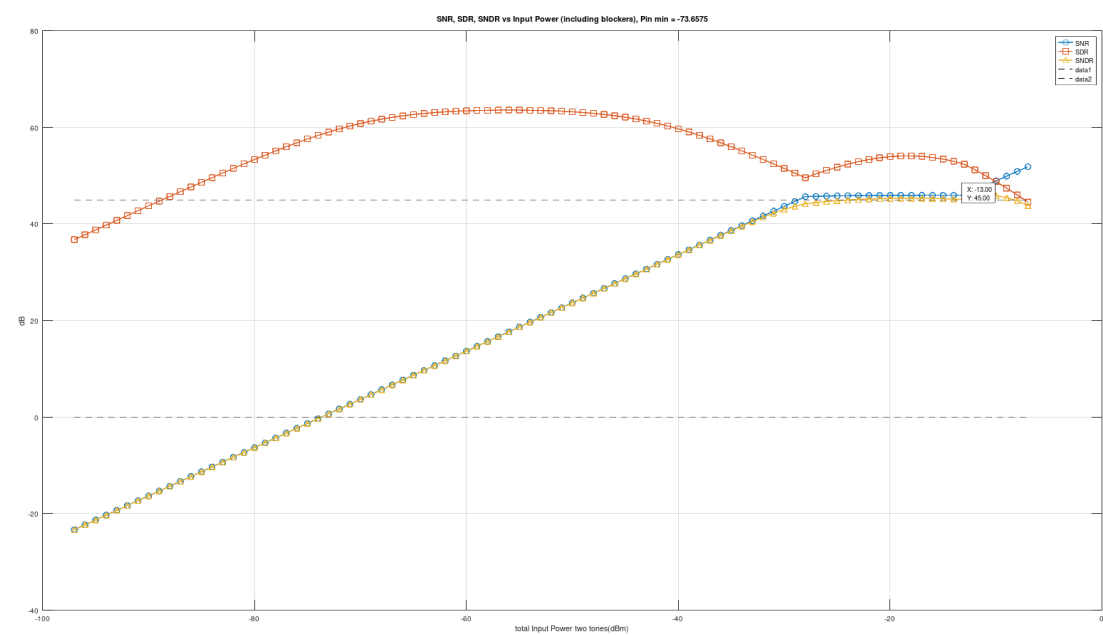
Original Lineup Results: Ex. + Int. LNAs (with blocker)



Original Lineup Results: Ex. LNA

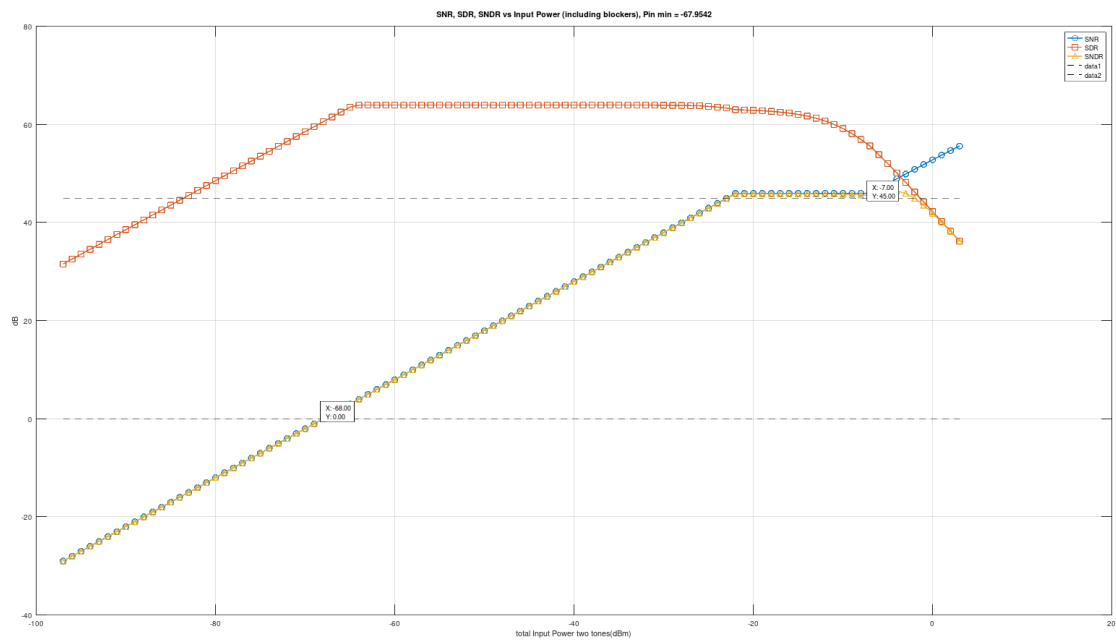


Without Blocker

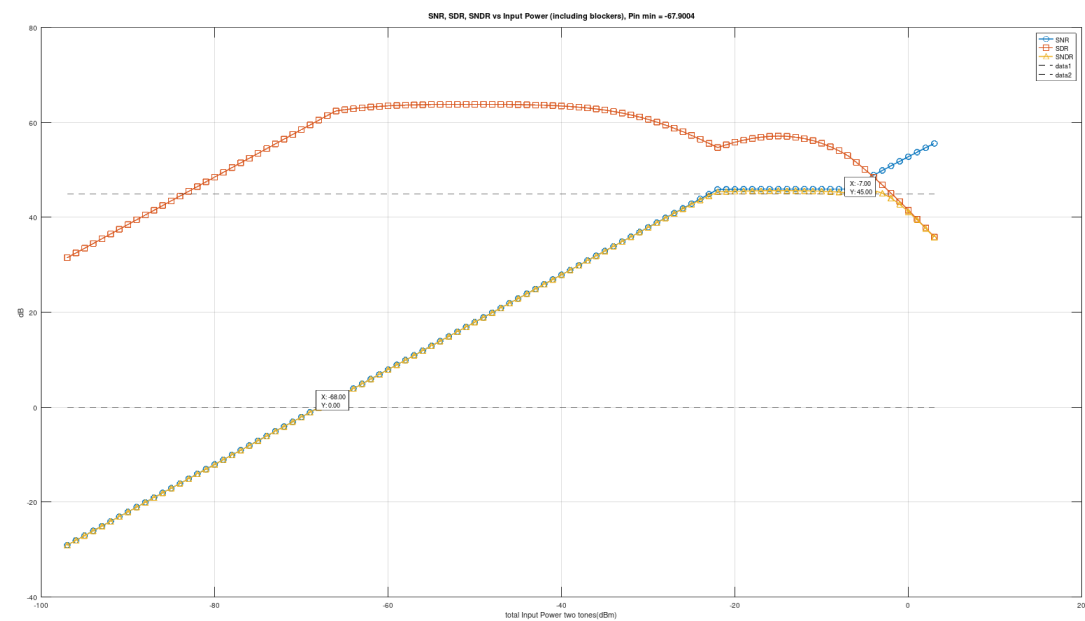


With Blocker

Original Lineup Results: Int. LNA



Without Blocker



With Blocker

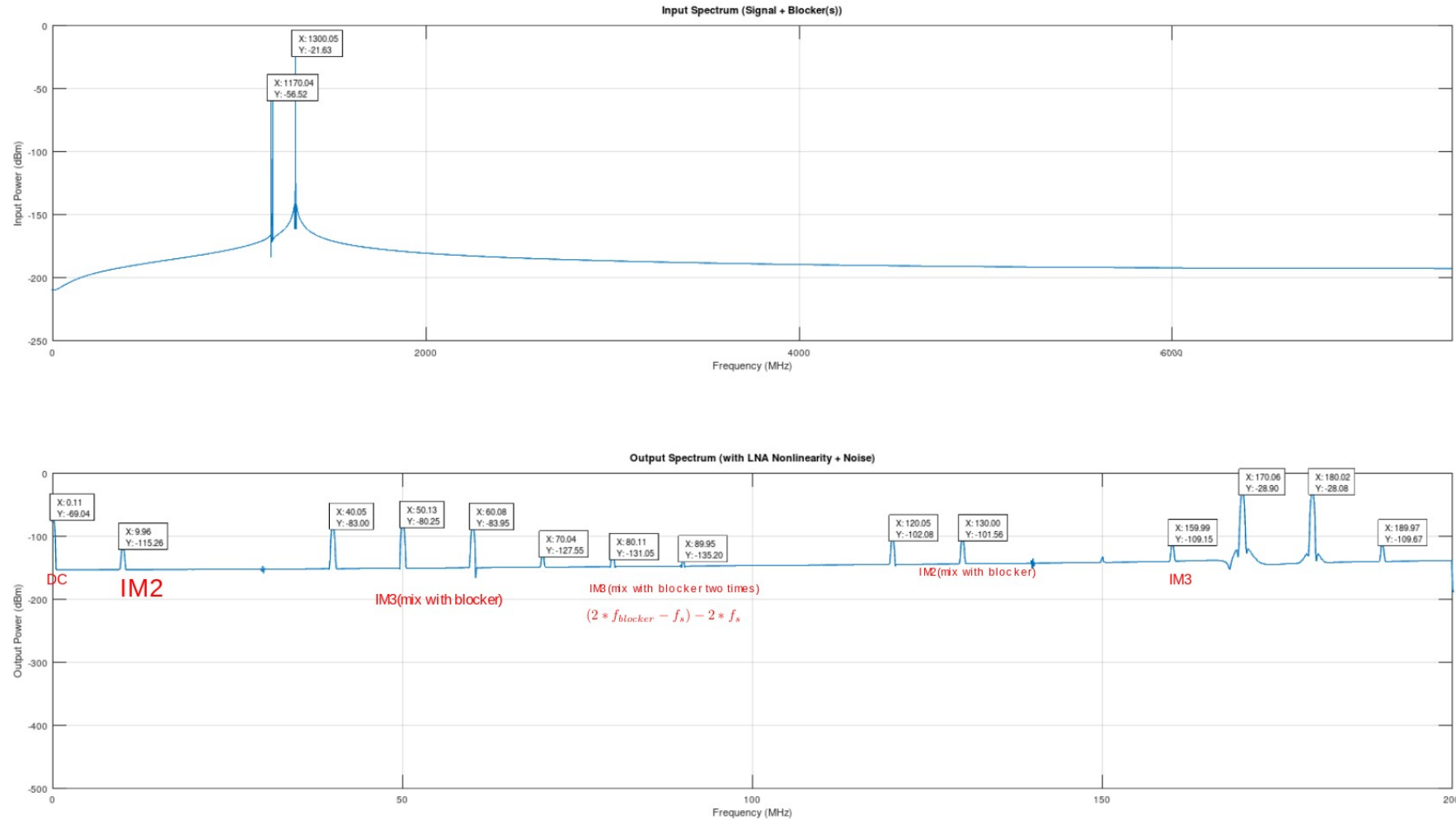
New Line-up: Assumptions

- Input signal :

Two tones at 1.17GHz and 1.18GHz

- Blocker: -15dBm at 1.3GHz
- LO : 1GHz
- LPF: BW = 200MHz , ideal filter

New Line-up: Output spectrum



New Line-up : Int. LNA (Design Equations):

$$\text{SDR due to blocker} = 2 \text{IIP3}_{\text{tillLPF}} - P_{\text{in}} - P_{\text{blocker}} - 3$$

$$\text{SDR without blockers} = 2(\text{IIP3} - P_{\text{in}})$$

Since Internal LNA linearity isn't so high we will only achieve $\text{SDR} = 45$ at $P_{\text{in}} = -20\text{dBm}$, in the case of No blockers exists, so IIP3 must be larger than -2.5dBm .

$$-91 + \text{NF} + G_{\text{max}} \geq -66 \text{ (ADC noise)}$$

$$\text{Assume } \text{NF} = 3, G_{\text{max}} \geq 22\text{dB}$$

$$\text{Gain Range} = 30\text{dB} \text{ (like the paper)}$$

$$G_{\text{Max}} = 27\text{dB}, G_{\text{min}} = -3\text{dB}$$

New line-up : Int. LNA line-up

System Description	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Component Name	LNA	Attenuator	Mixer	Filter	V G A
Gain (dB)	22	-3	-7	0	15
NF (dB)	2	3	7	5	10
P1dB (dBm)	13	15	6	6	10
IP3 (dBm)	23	25	16	16	20

OUTPUTS

Parameter	Output
Gain (dB)	27.00
NF (dB)	3.70
P1dB (dBm)	8.64
IP3 (dBm)	18.64

Line-up Max Gain

System Description	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
Component Name	LNA	Attenuator	Mixer	Filter	V G A
Gain (dB)	22	-23	-7	0	5
NF (dB)	2	23	7	5	10
P1dB (dBm)	13	15	6	6	10
IP3 (dBm)	23	25	16	16	10

OUTPUTS

Parameter	Output
Gain (dB)	-3.00
NF (dB)	18.94
P1dB (dBm)	-12.08
IP3 (dBm)	-2.32

Line-up Min gain

New line-up : Int. LNA line-up

We have added a VGA at the base band, to control to signal input to the ADC to be about -20 dBm on average.

- The range of the VGA is 10 dB, from 5dB to 15dB.
- The range of the DSA is 20dB, from -3dB to -23dB.

New line-up : Ext. LNA line-up

System Description Component Name	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
	LNA	Attenuator	Mixer	Filter	V G A
Gain (dB)	25	-23	-7	0	5
NF (dB)	2	23	7	5	10
P1dB (dBm)	13	15	6	6	10
IP3 (dBm)	35	25	16	16	10

OUTPUTS

Parameter	Output
Gain (dB)	0.00
NF (dB)	16.03
P1dB (dBm)	-12.08
IP3 (dBm)	6.56

Line-up Minimum Gain

System Description Component Name	Stage 1	Stage 2	Stage 3	Stage 4	Stage 5
	LNA	Attenuator	Select ...	Select ...	Select ...
Gain (dB)	25	-3	-7	0	15
NF (dB)	2	3	7	5	10
P1dB (dBm)	25	15	6	6	5
IP3 (dBm)	35	25	16	16	15

OUTPUTS

Parameter	Output
Gain (dB)	30.00
NF (dB)	2.94
P1dB (dBm)	4.71
IP3 (dBm)	14.71

Maximum Gain

New Line-up : Gain Policy:

- Two power detector is placed , one at the ouput to the ADC and the other after the mixer before LPF , this power detector gives information about the blocker existence.
- In case that that P_{mixer} is large enough that may degrade SDR , attenuation is done by DSA
- In case that $P_{\text{out}} > 20\text{dBm}$, Attenuation is done by VGA then DSA (if VGA reaches minmum)

$$\text{SDR}_{\text{due to blocker}} = 2 \text{IIP3}_{\text{till LPF}} - P_{\text{in}} - P_{\text{blocker}} - 3$$

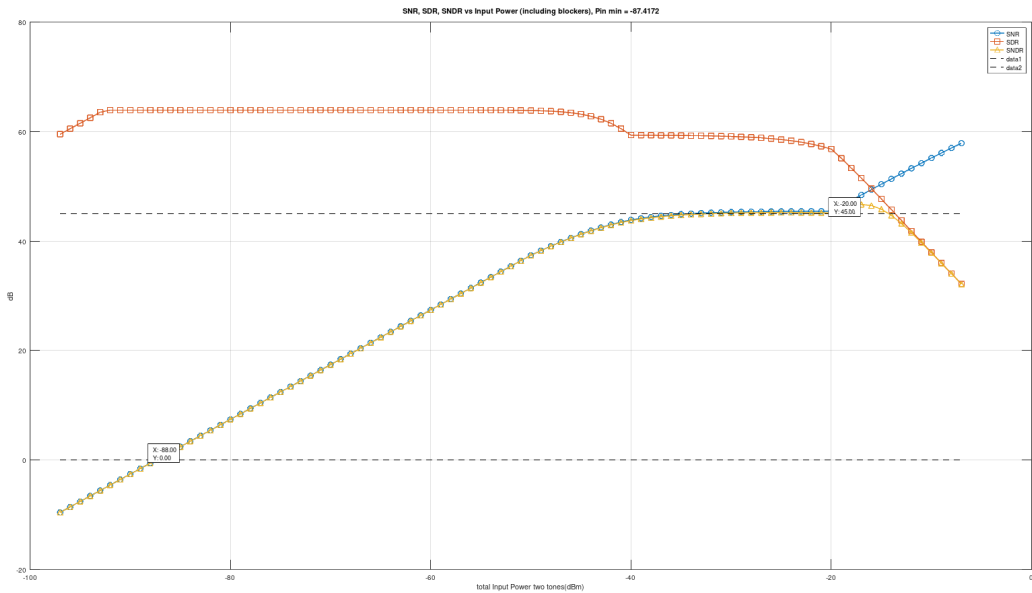
$$\text{SDR} = 2(\text{OIP3}_{\text{till LPF}} - G_{\text{total}} + G_{\text{VGA}}) - P_{\text{out}} + G_{\text{total}} - P_{\text{blocker}}$$

$$\text{SDR} = 2 \text{OIP3}_{\text{till LPF}} - (P_{\text{out}} + P_{\text{mixer}} + G_{\text{LPF}}) + G_{\text{VGA}}$$

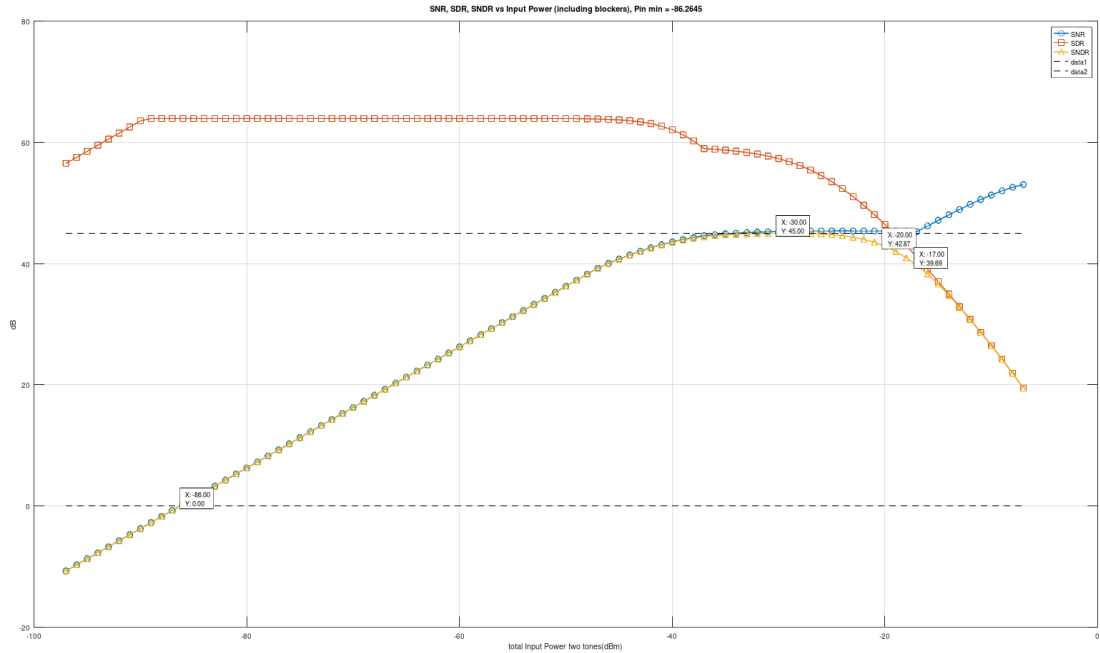
$$\text{let SDR} = 45\text{dB},$$

$$P_{\text{out}} + P_{\text{mixer}} \leq 2 \text{OIP3}_{\text{till LPF}} - 45 + G_{\text{AMP}} - G_{\text{LPF}}$$

SNR, SDR and SNDR without blockers

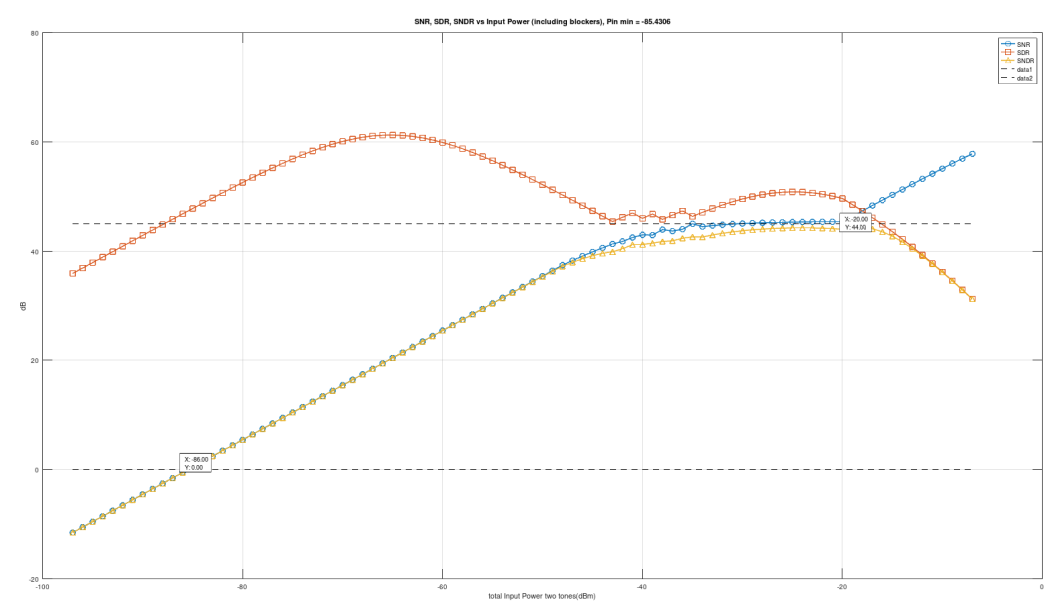


External LNA

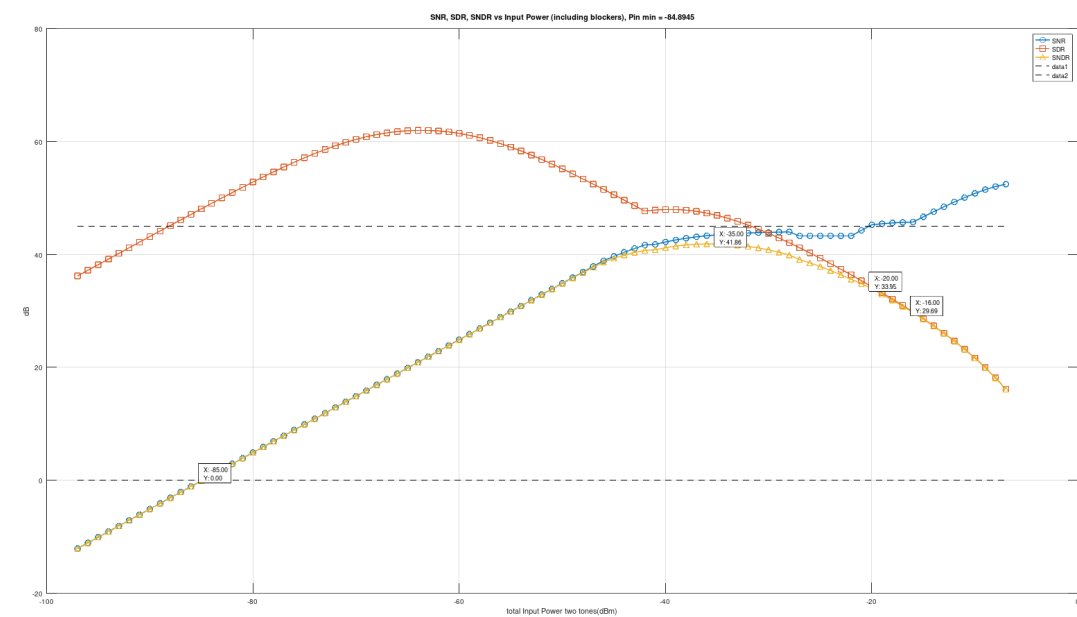


Internal LNA

SNR, SDR and SNDR with blockers

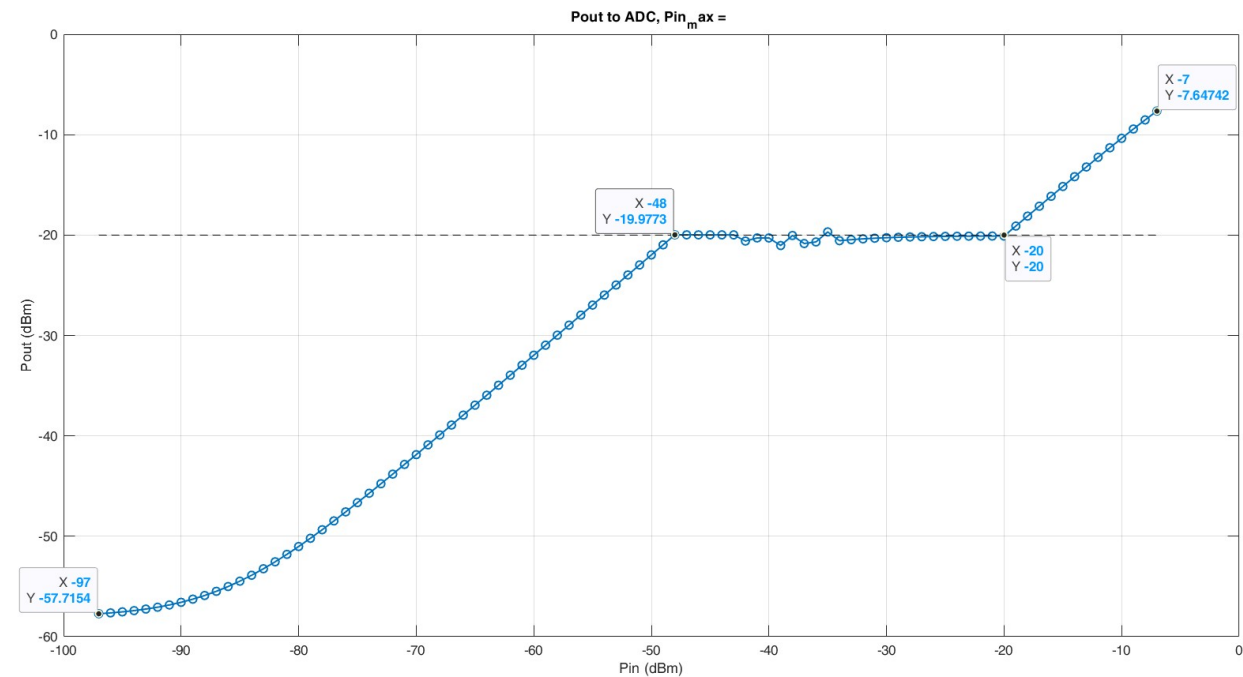


External LNA

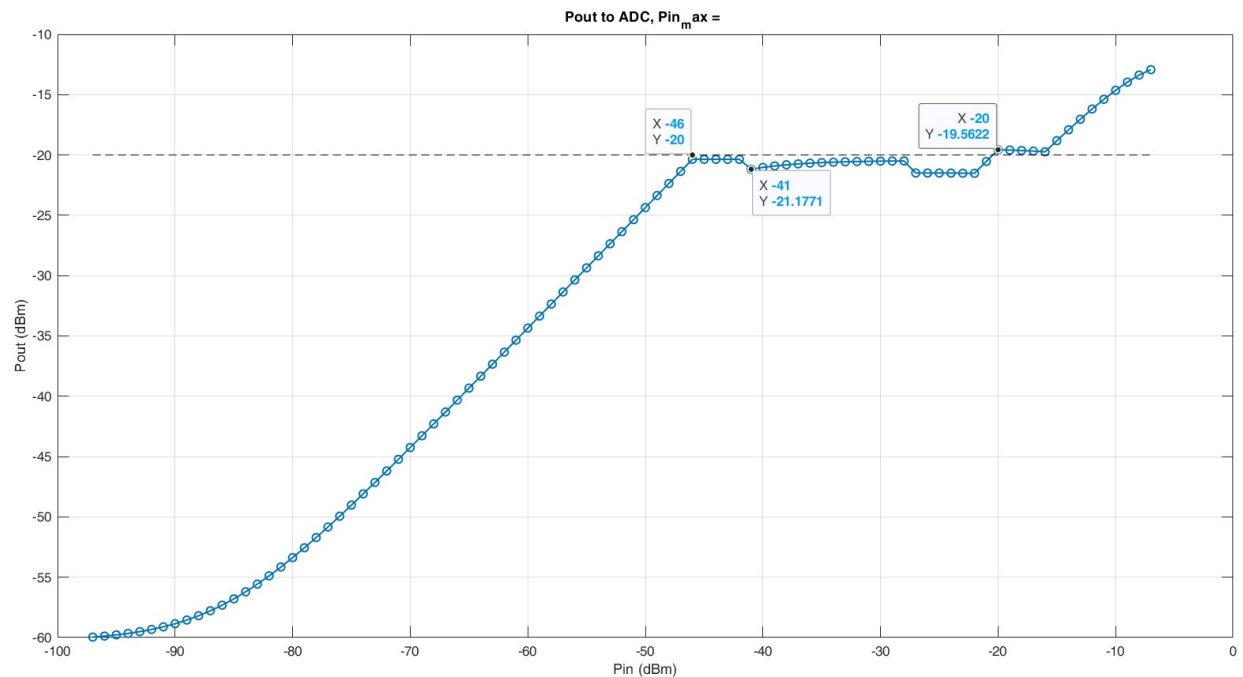


Internal LNA

Input power to ADC with blockers

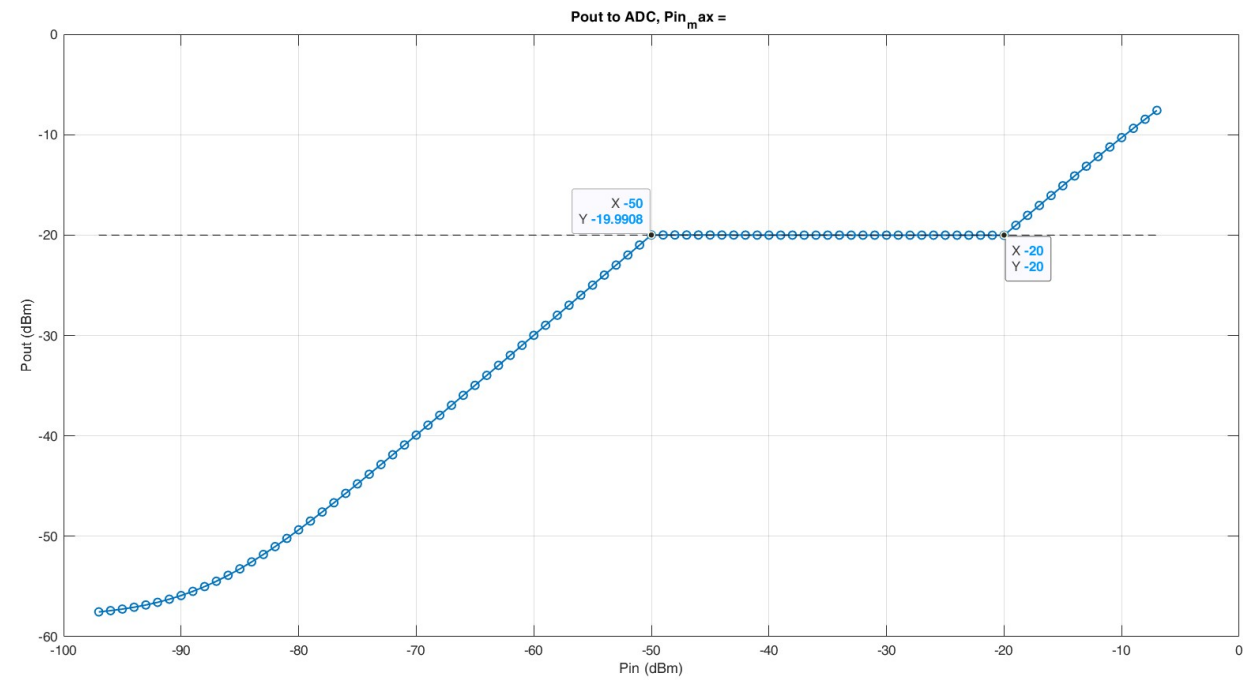


External LNA

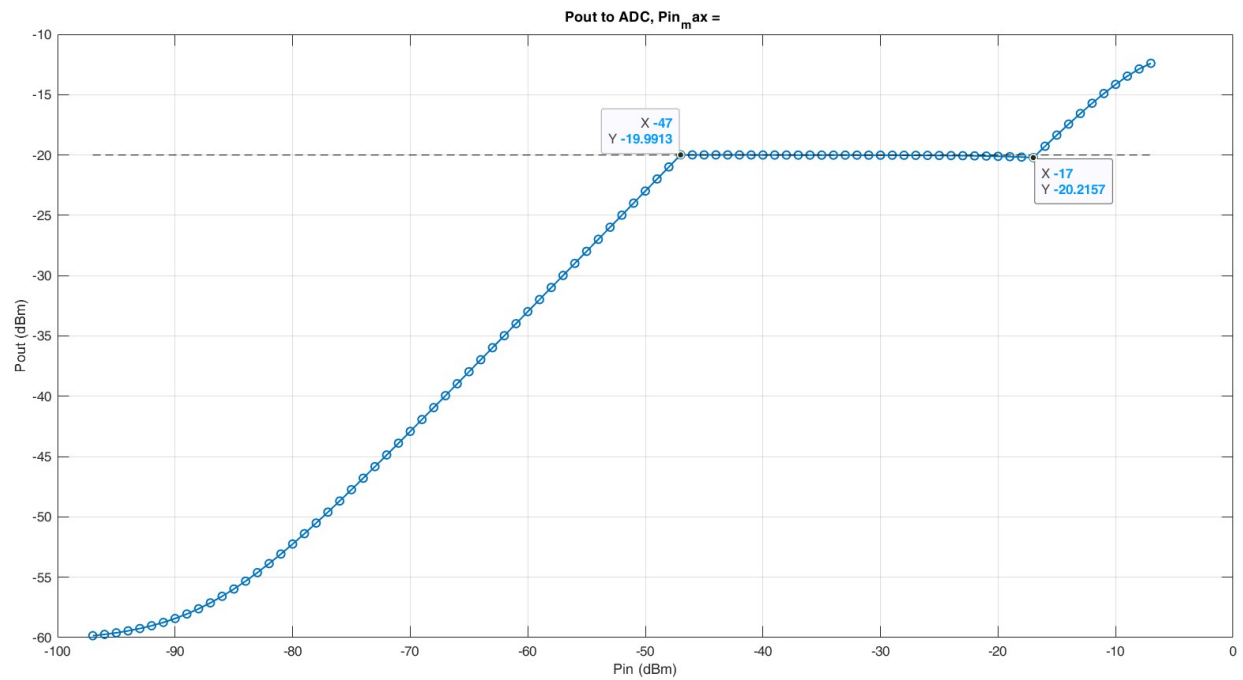


Internal LNA

Input power to ADC without blockers

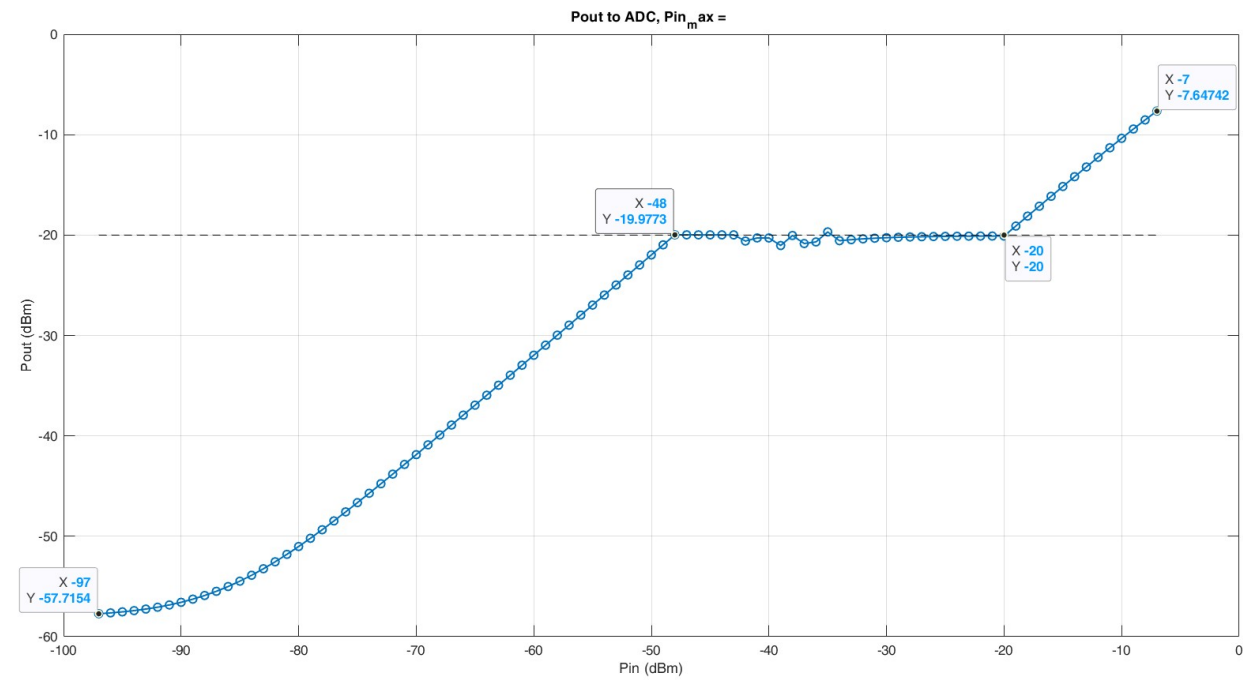


External LNA

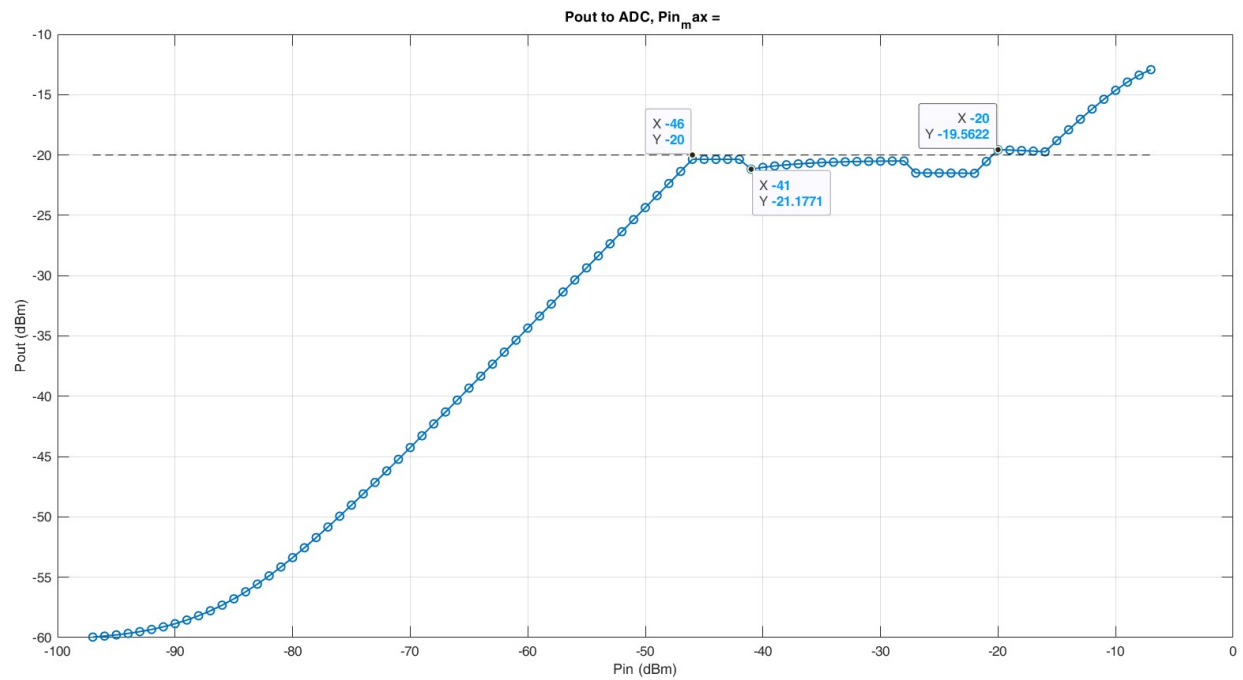


Internal LNA

Input power to ADC with blockers

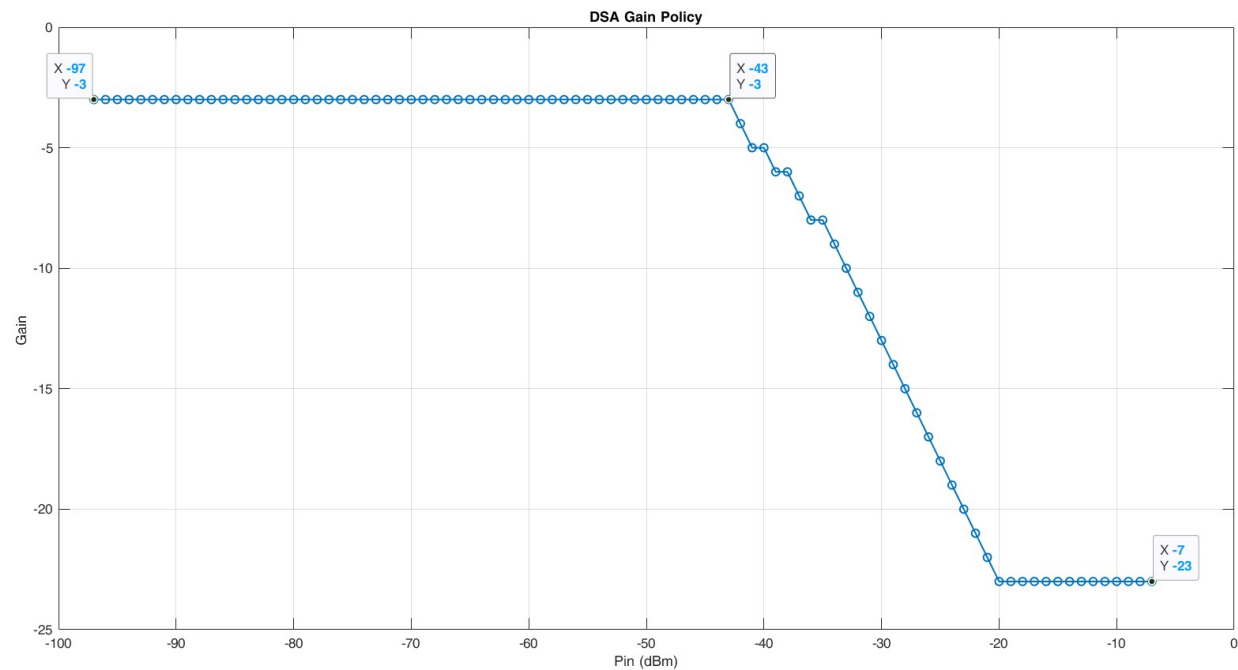


External LNA

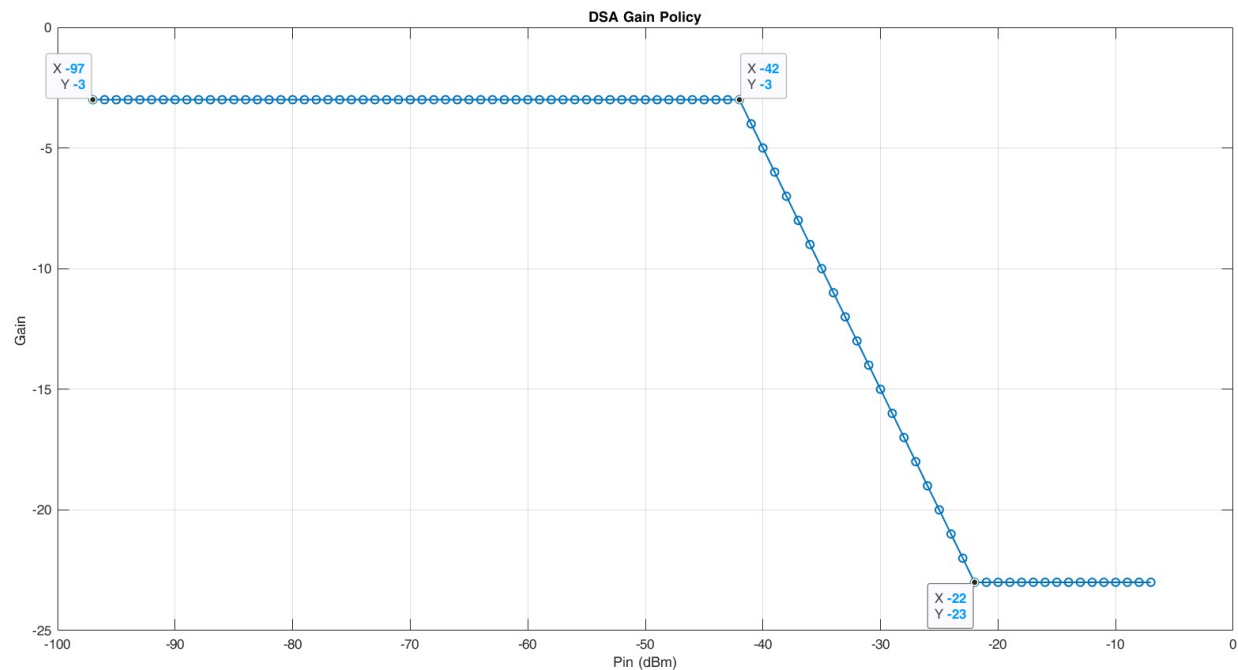


Internal LNA

DSA Policy with blockers

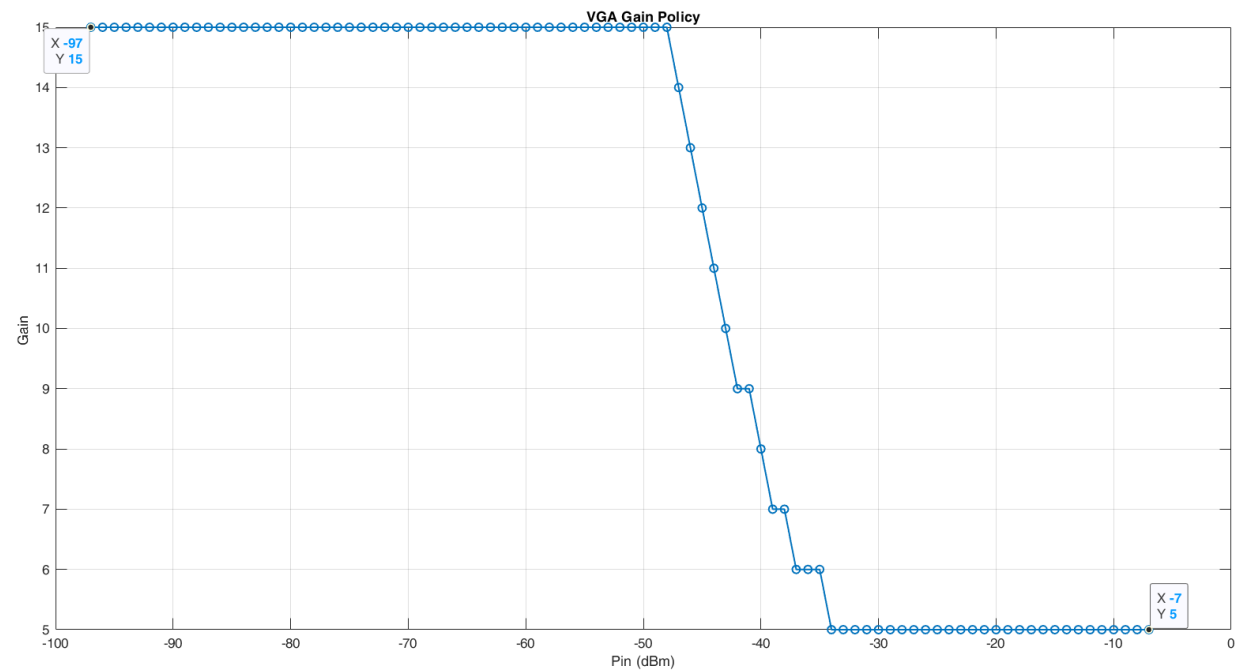


External LNA

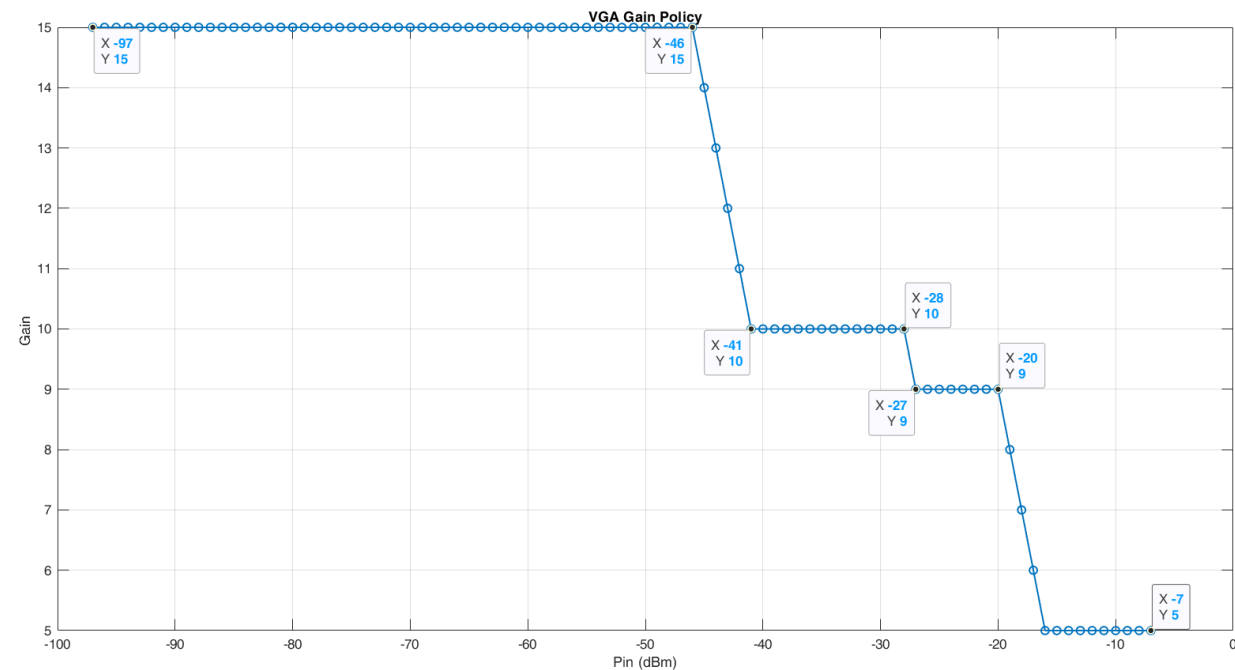


Internal LNA

VGA policy with blockers



External LNA



Internal LNA

Summary

	On chip LNA	External LNA	Notes
Gain	22dB	25dB	
OIP3	23dB	35dB	
SNDR $\geq$ 45dB (blockers)	Below 45dB	Below 45dB	Ex. LNA is better
SNDR $\geq$ 45dB (no blockers)	19 dBm	4 dBm	Ex. LNA is larger
Input power to ADC (no blockers)	30dBm range (-20dBm)	30dBm range (-20dBm)	Same performance
Input power to ADC (blockers)	28dBm range around -20dBm	26dBm range around -20dBm	Ex. LNA is better
SNR	-	-	For high input, Ex. LNA is better, but it doesn't matter as IP3 dominates

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Line-up Min gain

LPF Problem:

- 0.5dB BW = 250MHz
 - High rejection at 300 MHz ?
 - Two Modes of operation
 - OIP2 = 80dBm (out of band)
-
- IL = 0dB
 - OIP3 = 16dBm
 - NF = 5 dB

LPF	Parameter	Value	Unit
	0.5dB <u>VW</u>	250	MHz
	3dB <u>BW</u>	600	MHz
	Tunable filter	?	<u>TBD</u>
	In-band IL	2	dB
	OIP3	34	<u>dBm</u>

Original Specs